Ming Zhang

zhangming19@mails.ucas.ac.cn · (+86)15701204319
zhangming-19.github.io · GitHub · Google Scholar · ORCID

Self Introduction

- **Publications**: 6 SCI/SCIE indexed peer-reviewed articles (3 first author).
- **Academic impact**: Total citations: 28; H-index: 3; I10-index: 1 (as of March 19, 2024).
- **Areas of interests**: Natural Language Processing, Pre-trained Language Model, Fintech.
- **Hobbies & interests**: Enjoy reading, with a focus on classic works in economics.
  - Enjoys hiking and finds fulfillment in the sense of achievement that comes from trekking.
  - Enjoys swimming and often finds fresh perspectives and inspiration while swimming.

Education

University of Chinese Academy of Sciences — Beijng, China
School of Electronic, Electrical and Communication Engineering — 2019 - *present*
Signal and Information Processing • Doctor • GPA: 3.55/4.00

China University of Geosciences, Beijing — Beijng, China
School of Geophysics and Information Technology — 2015 - 2019
Measurement & Control Technology and Instrument • Bachelor • GPA: 3.78/4.00

Honors & Awards

Merit Student by University of Chinese Academy of Sciences	2020 & 2023
Excellent Student Cadre by University of Chinese Academy of Sciences	2020
Outstanding Graduates by the Ministry of Education, Beijing	2019
Merit Student by the Ministry of Education, Beijing	2019
May Fourth Advanced Individual by China University of Geosciences, Beijing	2019
Top Ten Undergraduate Students by China University of Geosciences, Beijing	2019
Excellent undergraduate student by China University of Geosciences, Beijing	2019
Excellent Student Cadre by China University of Geosciences, Beijing	2018
Merit Student by China University of Geosciences, Beijing	2016 & 2017 & 2018

Research Experience

Research on Multi-document Summarization Generation and Evaluation — 2022 - *present*

- Enhancing multi-document summarization with multi-granularity relationship-based extractor.
- Enhancing multi-document summarization through curriculum learning and mixture-of-experts.
- Enhancing automatic summarization evaluation using ROUGE combined with semantics.

Research on Large-scale Financial Text Analysis — 2020 - *present*

- Predicting short-term stock movements through sentiment analysis of massive news articles.
- Predicting long-term stock movements with fused textual features of Chinese research reports.

Research on Semantic Retrieval and Question Answering of News Data — 2022.06 - 2022.09

A sliver medal (5/379) of the semantic retrieval track in National Big Data and Computational Intelligence Challenge, 2022.

Research on Multi-level Emotion Recognition for Dialogue Data — 2020.09 - 2020.12

A sliver medal (3/75) of the emotion recognition track in Digital Grid Developer Competitions, 2020.

Selected Publications

International peer-reviewed journal articles ($N = 6$, first author = 3):

1. **Ming Zhang**, Jiyu Lu, Jiahao Yang, Jun Zhou, Meilin Wan, and Xuejun Zhang. From coarse to fine: Enhancing multi-document summarization with multi-granularity relationship-based extractor[J]. *Information Processing and Management*, 2024, 61(3): 103696. (SCI Q1, Rank: 11/158)
2. **Ming Zhang**, Chengzhang Li, Meilin Wan, Xuejun Zhang, and Qingwei Zhao. ROUGE-SEM: Better evaluation of summarization using ROUGE combined with semantics[J]. *Expert Systems with Applications*, 2024, 237: 121364. (SCI Q1)
3. **Ming Zhang**, Jiahao Yang, Meilin Wan, Xuejun Zhang, and Jun Zhou. Predicting long-term stock movements with fused textual features of Chinese research reports[J]. *Expert Systems with Applications*, 2022, 210: 118312. (SCI Q1)
4. Chengzhang Li, **Ming Zhang**, Xuejun Zhang, and Yonghong Yan. MCRSpell: A metric learning of correct representation for Chinese spelling correction[J]. *Expert Systems with Applications*, 2024, 237: 121513. (SCI Q1)
5. Jiahao Yang, **Ming Zhang**, Shuo Feng, Xuejun Zhang, and Xing Bai. A Hierarchical Deep Model Integrating Economic Facts to Predict Stock Movements, *Engineering Applications of Artificial Intelligence*, in press. (SCI Q1)
6. Jiahao Yang, Shuo Feng, Wenkai Zhang, **Ming Zhang**, Jun Zhou, and Pengyuan Zhang. An efficient loss function and deep learning approach for ranking stock returns in the absence of prior knowledge[J]. *Information Processing and Management*, 2024, 61(1): 103579. (SCI Q1)

Submitted papers

1. **Ming Zhang**, Jiyu Lu, Shuo Feng, Jiahao Yang, Jun Zhou, Meilin Wan, and Xuejun Zhang. Towards Curriculum Learning of Multi-Document Summarization using Difficulty-aware Mixture-of-Experts, first-round revision at *Knowledge-Based Systems*. (SCI Q1)
2. Jiyu Lu, **Ming Zhang**, Ta Li, and Yu Feng. Infant Cry Insight: Leveraging Enhanced ResNet and Data Augmentation for Classification, submitted to *Cogsci 2024*. (SCI)
3. Jiyu Lu, **Ming Zhang**, Ta Li, Yu Feng, and Xueshuai Zhang. Fan Fault Detection: Leveraging LSTM-CNNs with Acoustic Feature Fusion, submitted to *Interspeech 2024*. (EI)
4. Jiahao Yang, Ran Fang, **Ming Zhang**, Wenkai Zhang, and Jun Zhou. Separating the Predictable Part of Returns with CNN-GRU-Attention from Inputs to Predict Stock Returns, submitted to *Applied Soft Computing*. (SCI Q1)
5. Jiahao Yang, Wenkai Zhang, **Ming Zhang**, Jun Zhou, and Xuejun Zhang. Enhancing Stock Ranking Forecasting by Modeling Returns with Heteroscedastic Gaussian Distribution, submitted to *Expert Systems with Applications*. (SCI Q1)

Website development and maintenance

- Supplementary materials for "ROUGE-SEM: Better evaluation of summarization using ROUGE combined with semantics".
  Source: https://github.com/zhangming-19/ROUGE-SEM
- Supplementary materials for "Predicting long-term stock movements with fused textual features of Chinese research reports".
  Source: https://github.com/zhangming-19/SMPRR

Service to the profession

Referees for the following international peer-review journals:

Information Processing and Management	2024 - *present*
Expert Systems with Applications	2023 - *present*